

TrES-1b

R-1706

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_210805 · created 2026-08-02T14:59:41+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459431.7805 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1845 +/- 0.003
Transit depth (Rp/Rs)^2	3.4 +/- 0.11 [%]
Orbital Inclination (inc)	89.99 +/- 0.32
Airmass coefficient 1 (a1)	1.527 +/- 0.02
Airmass coefficient 2 (a2)	-0.412 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.32 %
Best Comparison Star	None
Optimal Aperture	4.42
Optimal Annulus	6.68
Transit Duration (day)	0.10868 +/- 0.00035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1845 ± 0.003 vs 0.1358 (deviation 12.19σ)
Duration fit vs published	0.10868 d vs 0.1042 d (deviation 9.68σ)
Mid-time O–C	-0.8 min
Depth SNR	46.2
Residual scatter	1.32 %

Light curve

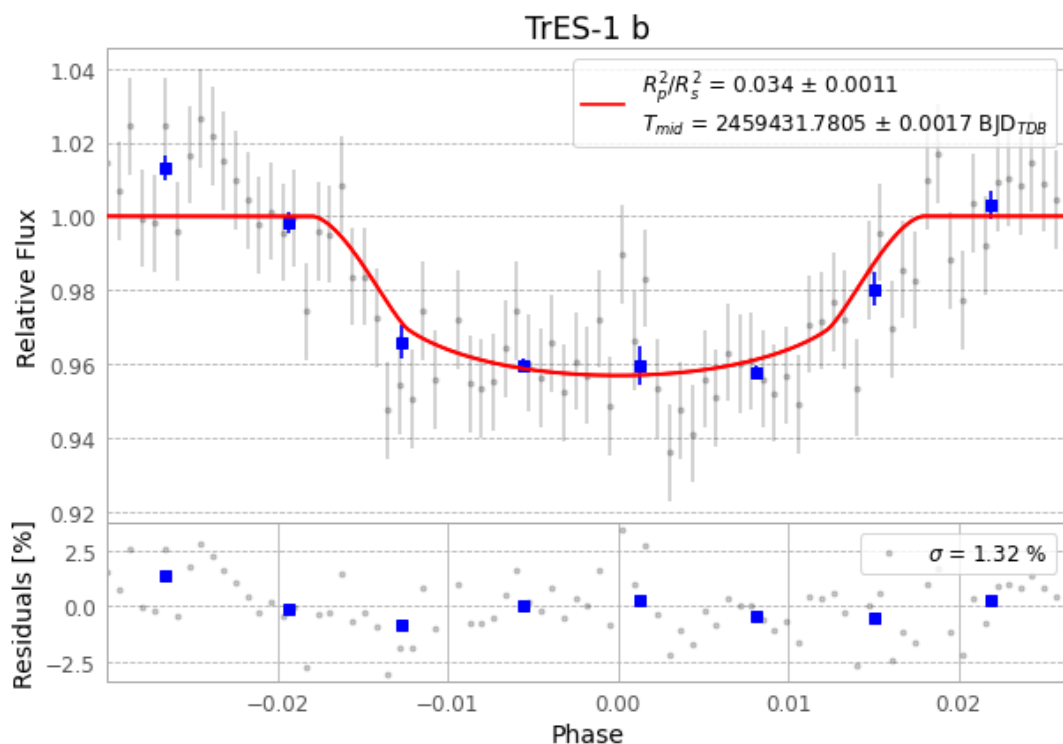


Figure 1 — FinalLightCurve_TrES-1 b_2021-08-04.png (SHA-256 c8ea93c4a796d74a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [388, 182], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f9865691727edf15...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), TRES-1210805042713.FITS → TRES-1210805083312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-210805.log (4da0fb574ecb...), FinalLightCurve_TrES-1 b_2021-08-04.png (c8ea93c4a796...), FinalLightCurve_TrES-1 b_2021-08-04.csv (284ded30b223...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 14:59Z.